

Answer ALL questions.

Write your answers in the spaces provided.

- 1 Squid is an animal that is eaten in many parts of the world.

The body of the squid is used as meat because it contains a lot of protein.

The photograph shows some squid meat.



(Source: © Ilia Nesolenyi/Alamy Stock Photo)

The meat from squid is quite tough and is often treated with enzymes to break down the protein.

Some fruits contain enzymes that can break down protein.

Pineapples contain the enzyme bromelain and unripe papayas contain the enzyme papain.

A study compared the effects of two different concentrations of these two enzymes on the meat from squid.

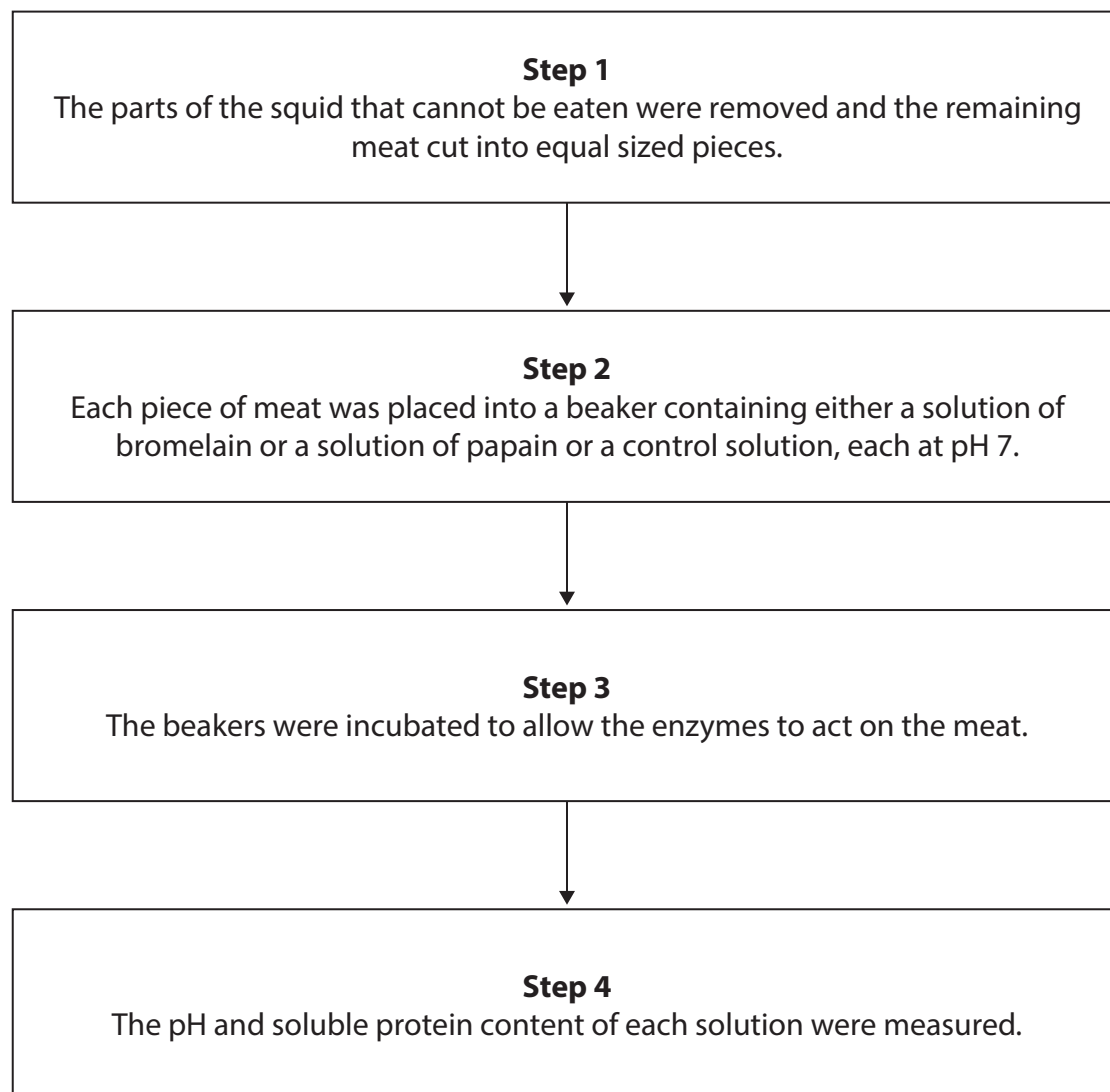
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



The flowchart shows some of the steps in the method used in this study.



- (a) Explain why, in **step 1**, it is important to cut the squid meat into equal sized pieces.

(2)

- (b) (i) Name a suitable control solution that could be used in **step 2**.

(1)

- (ii) State the importance of using a control in this study.

(1)

- (c) Give **two** variables that need to be controlled in **step 3**.

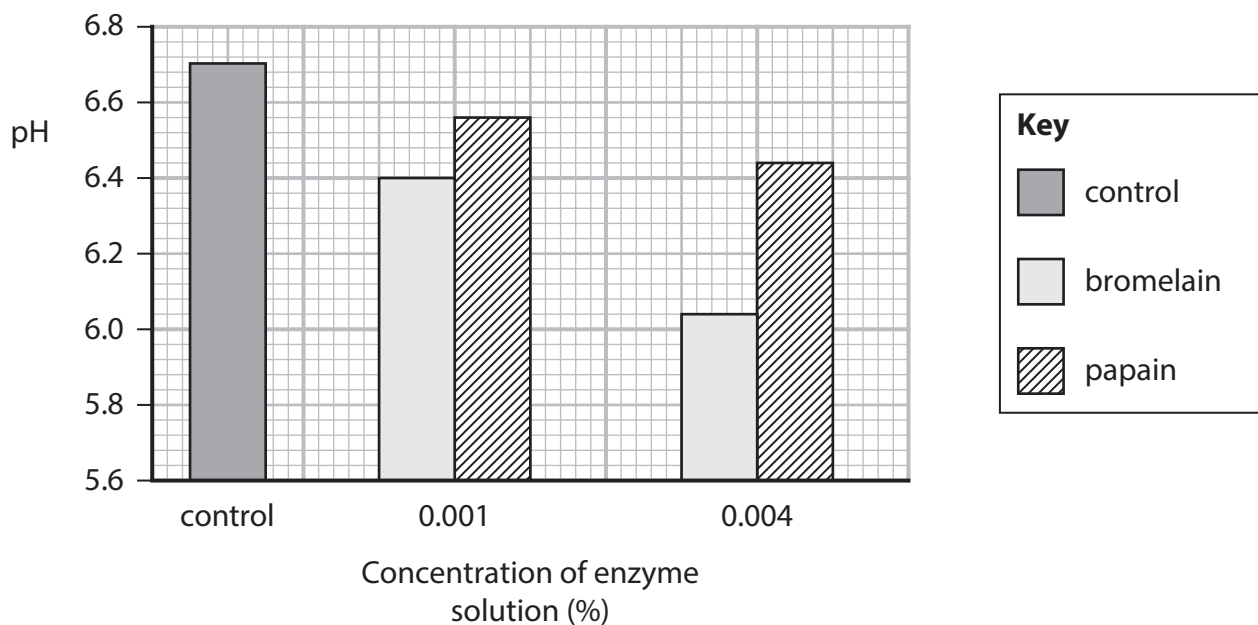
(1)

1

2



(d) The graph shows the pH of the solutions measured at **step 4**.



- (i) The 0.001% enzyme solution was prepared by diluting the 0.004% enzyme solution.

Calculate the volume of the 0.004% enzyme solution needed to make 50 cm³ of 0.001% enzyme solution.

(1)

Answer cm³

- (ii) State **two** conclusions that can be made from the data shown in the graph.

(2)

1

.....

.....

2

.....

.....



P 7 5 7 7 3 A 0 5 2 0

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (e) Describe a method for estimating the soluble protein concentration in the beakers in **step 4**.

(4)

(Total for Question 1 = 12 marks)

